1. Abstract:

The design, hardware implementation, and performance analysis of pulse position modulation (PPM), pulse width modulation (PWM), and pulse amplitude modulation (PAM) systems, as well as an passive Low-Pass Filter (LPF) demodulator, are presented in this project. Operating under strict laboratory voltage constraints ($\leq \pm 10$V DC) and low-frequency signal ranges (50 Hz $< f_m <$ 95 Hz), the system successfully generates and reconstructs modulated signals without distortion or over-modulation. This configuration is designed as a standardized, low-power educational laboratory module that goes beyond technical validation. It connects theoretical concepts of digital communication with actual laboratory application by allowing junior undergraduate students to practically view, analyze, and compare the mechanics of pulse modulation schemes on oscilloscopes.

2. Objectives

- To design and construct reliable hardware circuits for PAM, PWM, and PPM generation under bounded DC supply voltage limits ($\leq \pm 10$V DC).
- To design an passive Low-Pass Filter (LPF) demodulator to accurately recover the original analog message signal.
- To experimentally verify theoretical frequency relationships (Nyquist criteria and carrier thresholds) and demonstrate over-modulation prevention ($V_m < V_c$).

3. Introduction

Pulse modulation techniques form the foundation of modern digital communication and signal processing systems. In analog pulse modulation schemes, a continuous-time analog message signal modifies specific properties of a periodic pulse train carrier to encode information. Among these schemes, Pulse Amplitude Modulation (PAM) operates by varying the amplitude of the carrier pulses in direct proportion to the instantaneous amplitude of the message signal while keeping pulse duration and position constant. Pulse Width Modulation (PWM) alters the width or duration of the pulses proportionally to the message signal while maintaining a constant pulse height, thereby providing improved noise immunity and power efficiency. Pulse Position Modulation (PPM) adjusts the relative timing position or phase shift of fixed-width pulses based on variations in the message signal, optimizing average power transmission efficiency.

In practical electronic communication laboratories, implementing these modulation schemes involves strict operational boundary conditions and frequency relationships to ensure accurate signal sampling and transmission. Demodulation across all three schemes is achieved using active Low-Pass Filters, which eliminate high-frequency pulse train harmonics and isolate the baseband message signal. Beyond theoretical validation, establishing functional hardware setups for PAM, PWM, and PPM provides an essential pedagogical tool, allowing

engineering students to visually compare and analyze continuous-to-discrete modulation dynamics on oscilloscopes.

4. Methodology:

The methodology adopted for this project follows a comprehensive engineering lifecycle—from initial circuit calculations to complete PCB trainer kit fabrication:

- Phase 1 (System Architecture & Calculation): A modular architecture was designed where a low-frequency sine wave (50 Hz < fm <95 Hz) acts as the message input. Switching and pulse generation circuits produce carrier signals satisfying specific frequency limits.
- Phase 2 (PAM Stage Design): Implemented using a CMOS Analog Switch (CD4016) driven by a 1.5 kHz square wave clock signal.
- Phase 3 (PWM Stage Design): Constructed using an Op-Amp Comparator (LM358) comparing the message signal with an 2 kHz sawtooth carrier.
- Phase 4 (PPM & Inverting Stage Design): Developed using an Op-Amp-based Monostable Multivibrator triggered by PWM falling edges. An additional Op-Amp Inverting Amplifier (Av = 1) stage was integrated to invert the negative-going pulses back to positive polarity.
- Phase 5 (Active Demodulator Design): Designed using a $3^{rd}/4^{th}$ -order passive Low-Pass Filter (fc ≈ 100 Hz) for smooth sine wave reconstruction.
- Phase 6 (Breadboard Prototyping & Testing): Circuits were initially assembled on a breadboard to verify operational parameters, component values, and signal waveform integrity under Dual Supply <= +/-10V DC constraints.
- Phase 7 (PCB Design & Fabrication): Following breadboard validation, the schematic was transferred to PCB layout software for track routing, ground plane optimization, and soldering component mounting.
- Phase 8 (Trainer Kit Enclosure Integration): The finalized PCB board, power terminals, input/output test points, and control knobs were mounted inside a protective enclosure box, creating an integrated, standalone laboratory trainer kit.

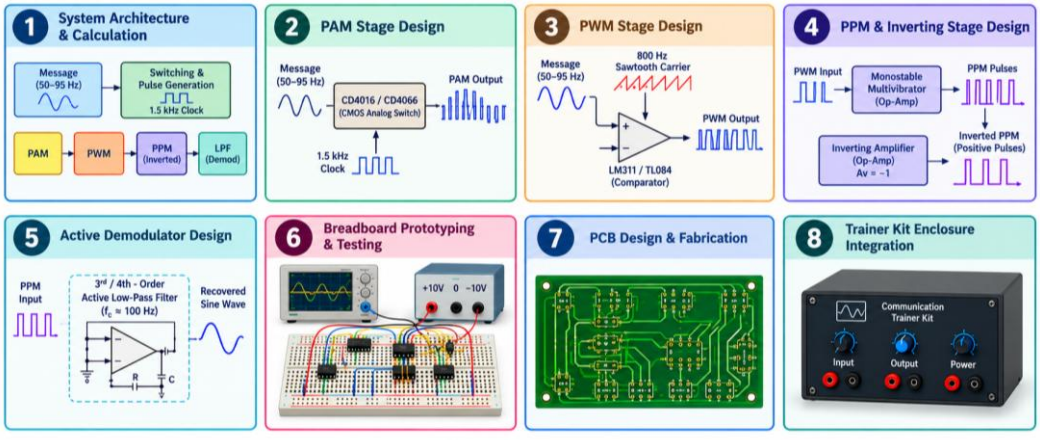


Figure 1: System flowchart of PAM, PWM, PPM Modulation Kit

5. Apparatus Required

Table 1: Component requirement table

Sl. No.	Equipment	Specification / Type	Quantity
1	Dual DC Power Supply	Regulated (+/-10V DC Max)	1 Unit
2	Signal / Function Generator	Sinusoidal & Triangular/Sawtooth Wave	2 Units
3	Digital Oscilloscope	20 MHz / Dual Channel	1 Unit
4	Operational Amplifiers	UA741, LM358	3 ICs
5	Bilateral Analog Switch IC	CD 4016	1
5	Resistors	1k, 10k, 100k	As Required
6	Capacitors	0.1uF, 0.01uF	As Required
7	Breadboard & Connecting Wires	Standard Lab Setup	1 Set

6. Software Requirement:

Table 2: Software requirement table

S.L No	Software Name
1	Proteus 8 professional
2	EdrawMax

7. System Parameters & Design Data Table(recommended)

Table 3: Recommended system parameter

Modulation Type	Message Voltage (Vm)	Message Freq (fm)	Carrier Voltage (Vc)	Carrier Freq (f_c)	Operating Power Rail
PAM	$V_m < V_c$ (e.g., 6V - 8V p-p)	50 Hz < f_m < 95 Hz (e.g., 90 Hz)	V_c = Supply Voltage (+/- 10V or 10V p-p)	f_c > 1 kHz (e.g., 1.5 kHz)	Dual Supply ≤ 10V (DC)
PWM	$V_m < V_c$ (e.g., 5V - 7V p-p)	50 Hz < f_m < 95 Hz (e.g., 90 Hz)	Vc ≤ Supply Voltage (9V - 10V p-p Traingular)	f_c > 1.5 kHz (e.g., 2 kHz)	Dual Supply ≤ 10V (DC)
PPM	Derived from PWM (Vm < Vc)	50 Hz < f_m < 95 Hz (e.g., 90 Hz)	Vc ≤ Supply Voltage (9V - 10V p-p Traingular)	400 Hz < fc < 1 kHz (e.g., 600 Hz)	Dual Supply ≤ 10V (DC)

8. Circuit Diagram:

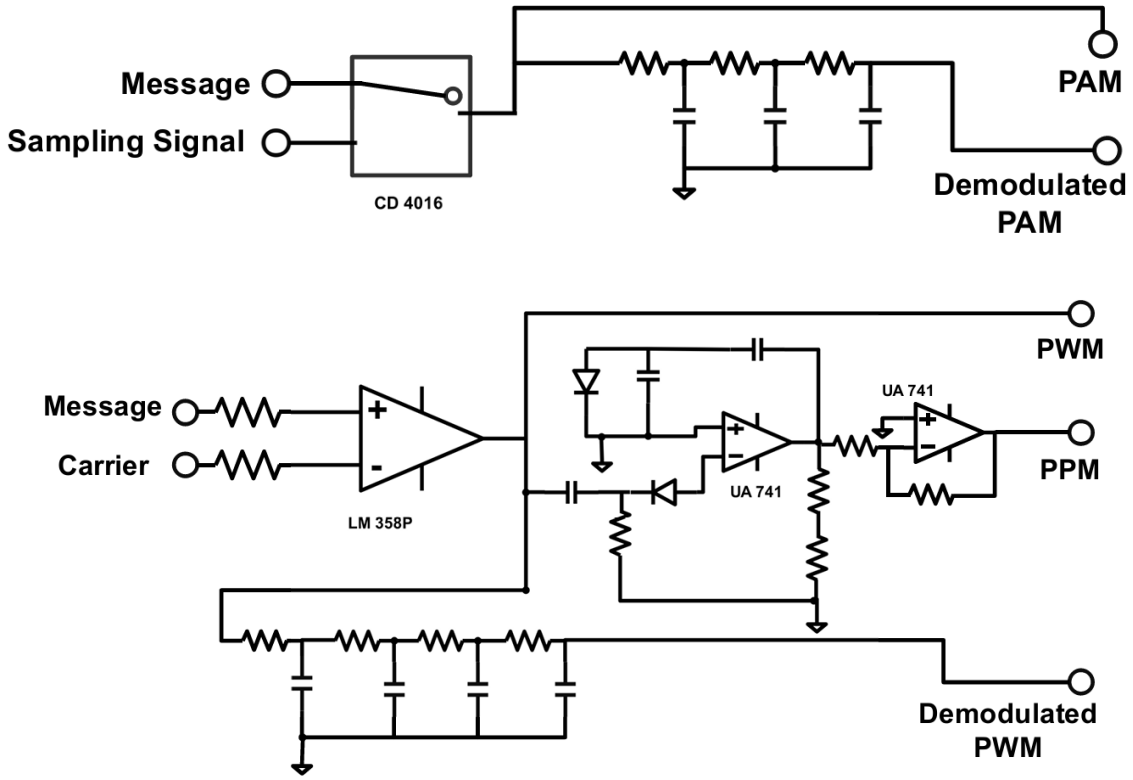


Figure 2: Circuit Diagram of PAM, PWM, PPM Modulation Kit

9. System Component Description:

a. CD4016 / CD4066 — CMOS Analog Switch:

Application: Pulse Amplitude Modulation (PAM) Generation

The CD4016 or CD4066 CMOS analog switch is used as the sampling element in the PAM generation stage. A periodic clock signal with a frequency above 2 kHz is applied to the control terminal to switch the analog input on and off at regular intervals. This process samples the low-frequency message signal and produces a PAM waveform whose pulse amplitudes follow the instantaneous values of the input signal.

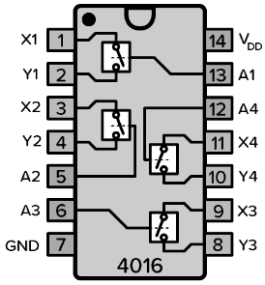


Figure 3: Pinout of DC4016

b. LM358 — Comparator:

Application: Pulse Width Modulation (PWM) Generation

The LM358 comparator ic is used to generate PWM signals. It compares the instantaneous amplitude of the low-frequency message signal (50–95 Hz) with a high-frequency sawtooth carrier (1.5 kHz– 2 kHz). The comparison produces a pulse train with a variable duty cycle corresponding to the message signal.

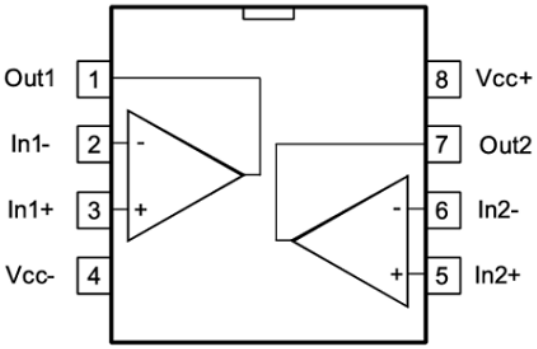


Figure 4: Pinout of LM358

c. UA741 — Operational Amplifiers:

Applications: PPM Generation, Pulse Inversion

Operational amplifiers are used in several stages of the communication system:

- Monostable Multivibrator: Generates fixed-width pulses triggered by the falling edges of the PWM signal, producing a PPM waveform.
- Unity-Gain Inverting Amplifier: Uses an inverting configuration with a voltage gain of (Av = 1) to restore the required pulse polarity.

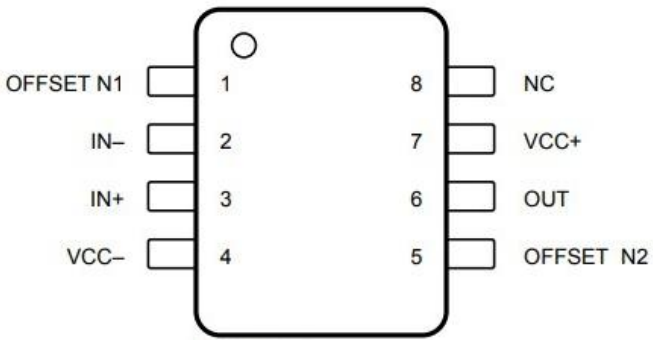


Figure 5: Pinout of UA741

d. Diode:

Application: PWM Edge Detection

The 1N4007 switching diode is incorporated into the monostable trigger network to provide edge-selective triggering. With the appropriate diode orientation and biasing arrangement, it helps suppress unwanted transitions and allows the negative-going trigger pulses associated with the falling edges of the PWM signal to activate the monostable circuit.

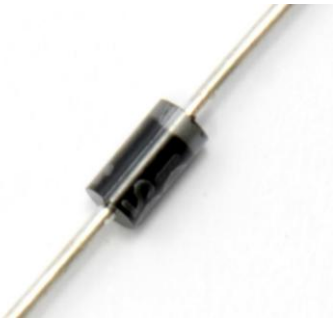


Figure 6: 1N4007 Diode

e. Resistors:

Resistors are used throughout the circuit for signal conditioning, gain adjustment, feedback, current limiting, and frequency-response control.

- Gain Setting: Equal input and feedback resistances of 10 kΩ are used in the unity-gain inverting amplifier to obtain a voltage gain of approximately -1.
- Feedback Network: Resistors establish the required feedback and switching thresholds in the monostable circuit.
- Filter Design: Resistor values work with the filter capacitors to determine the cutoff frequency of the active low-pass filter.



Figure 7: Resistor

f. Capacitors:

Non-polarized plastic capacitors are used in different stages of the communication system for pulse timing, signal filtering, and power-supply noise reduction. These capacitors can be connected without regard to polarity and are suitable for the timing and signal-processing applications used in this project.

- Timing Network: Used in the op-amp monostable multivibrator circuit to establish the RC timing characteristics and control the pulse duration of the PPM signal.
- Active Low-Pass Filter: Used in the third and fourth-order Sallen–Key low-pass filter. Along with the associated resistors, these capacitors determine the filter cutoff frequency, designed to be approximately 100 Hz, for reconstructing the original 50–95 Hz message signal.
- Power-Supply Decoupling: Connected near the IC supply pins to suppress high-frequency noise and improve the stability of the circuit.



Figure 8: Non-polarized Capacitor

g. Custom-Fabricated Printed Circuit Board (PCB):

The custom PCB integrates the PAM, PWM, PPM, and active low-pass filter circuits into a compact assembly. It provides organized component placement, reliable electrical connections, and shorter signal paths compared with temporary breadboard wiring. Proper PCB layout and grounding also help reduce unwanted noise and parasitic effects.

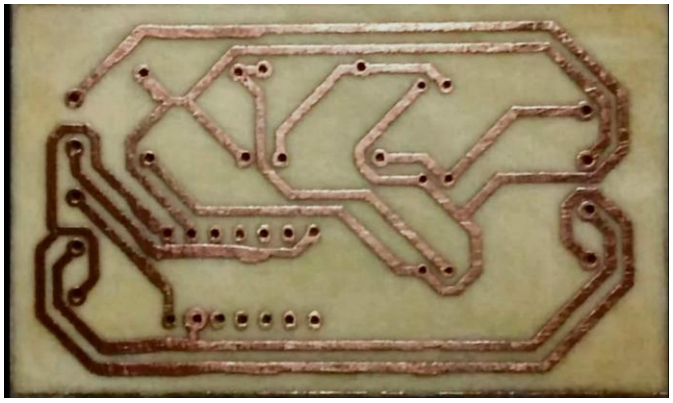


Figure 9: Printed PCB board of PAM

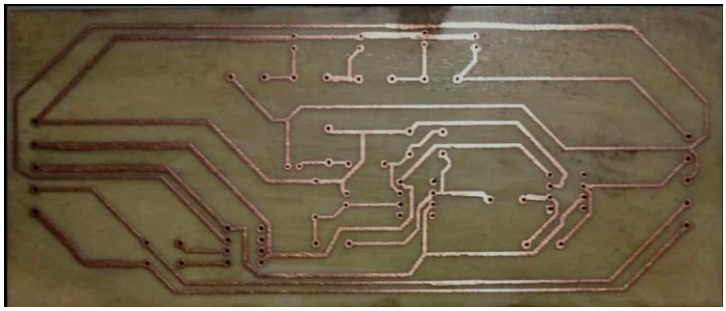


Figure 10: Printed PCB board of PWM & PPM

h. Project Enclosure Box:

The enclosure houses the completed PCB, power-supply connections, and front-panel interfaces. It protects the electronics from accidental contact and mechanical damage while providing a compact, portable, and organized laboratory training platform.

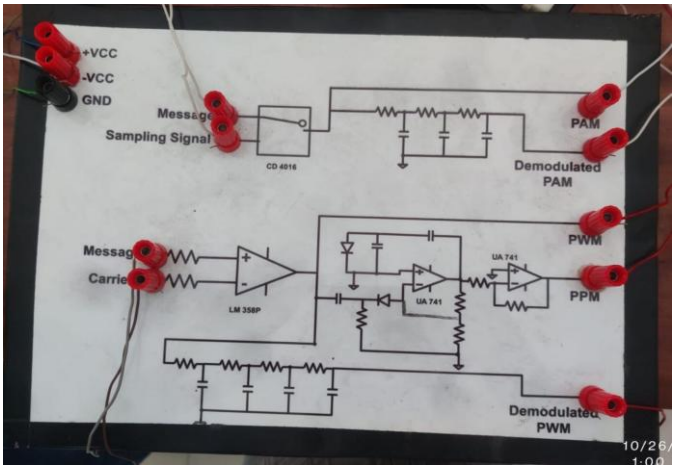


Figure 11: Top view of modulation kit

i. Hardware Implementation:

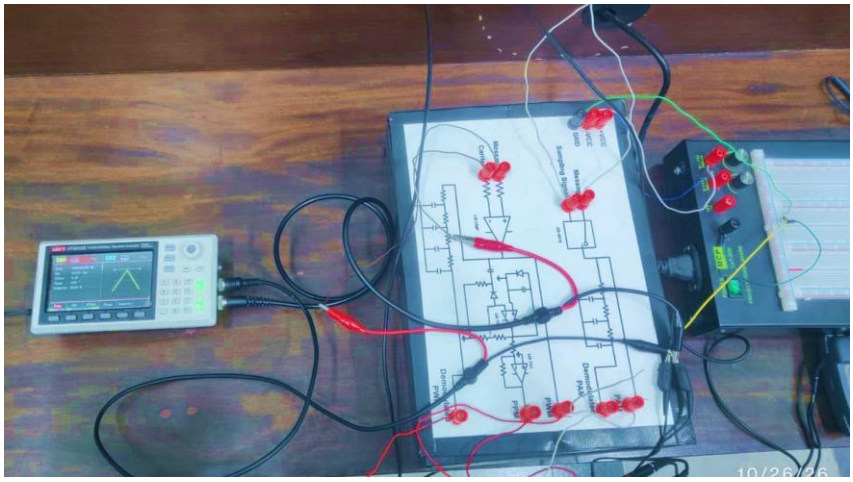


Figure 12: Hardware implementation of PAM, PWM, PPM Modulation Kit

10. Working Principle:

a. Signal Generation and Power Supply Initialization:

The communication trainer kit begins its operation by supplying the required message and carrier signals to the respective modulation stages. A function generator provides a low-frequency sinusoidal message signal within the range of (50<fm<95Hz}), selected as a representative operating frequency. The carrier signals required for PAM, PWM, and PPM generation are established according to their respective operating requirements:

- **PAM:** A sampling clock with a frequency above 1kHz, typically 1.5 kHz.
- **PWM:** A pulse-generation or timing signal with a frequency above 1.5 kHz, where required by the selected circuit configuration.
- **PPM:** A sawtooth carrier within the range of (300 Hz<fc<1 kHz), typically(60000Hz).

b. Pulse Amplitude Modulation (PAM):

The PAM stage uses a CD4016 analog switch to sample the continuous sinusoidal message signal at regular intervals. A (1.5 kHz) square-wave clock is applied to the switch's control terminal to determine the sampling instants. When the clock activates the switch, the instantaneous voltage of the message signal passes through to the output. When the switch is inactive, the signal path is interrupted, producing intervals between successive pulses. Consequently, the output consists of discrete pulses whose amplitudes follow the instantaneous values of the original sinusoidal message signal. This waveform is known as Pulse Amplitude Modulation (PAM) because the pulse amplitudes vary with the message signal, while the sampling clock determines their timing.

c. Pulse Width Modulation (PWM):

The PWM stage employs an LM358 comparator circuit. The sinusoidal message signal is applied to the non-inverting (+) input, while a sawtooth/triangular carrier is applied to the inverting (-) input. The comparator continuously compares the instantaneous voltages of these two signals. When the message voltage exceeds the carrier voltage, the output switches to its HIGH state. When the carrier voltage exceeds the message voltage, the output switches to its LOW state. As the message signal varies, the crossing points between the sine wave and the sawtooth carrier shift in time. This changes the duration of the HIGH and LOW intervals, producing rectangular pulses with a variable duty cycle. The pulse repetition rate is primarily determined by the carrier frequency, while the pulse width varies according to the message signal amplitude. Thus, the PWM waveform represents the information contained in the original message through variations in pulse duration rather than pulse amplitude.

d. Pulse Position Modulation (PPM) and Pulse Polarity Inversion:

The PPM generation stage converts the PWM signal into a pulse train with nearly constant pulse width and varying pulse positions. An op-amp-based monostable multivibrator is used to perform this conversion by responding to the falling edges of the incoming PWM signal. Whenever a falling edge of the PWM waveform occurs, it triggers the monostable circuit and generates a pulse of predetermined duration. Since the falling edges shift in time according to the instantaneous amplitude of the original message signal, the generated pulses also shift their positions along the time axis. In this way, the information carried by the message signal is represented by the relative timing of the output pulses rather than by their amplitude or width. Following the monostable stage, a unity-gain inverting amplifier is incorporated to correct the pulse polarity. With an appropriate resistor configuration, the amplifier provides a voltage gain of (Av=1), inverting negative-going pulses into positive-going pulses while ideally maintaining their amplitude magnitude and timing characteristics. The final output is a positive-polarity PPM waveform consisting of narrow pulses with approximately constant width and message-dependent pulse positions. This waveform demonstrates the fundamental principle of PPM and provides the required signal for subsequent observation and analysis in the communication trainer kit.

e. Demodulation and Baseband Signal Reconstruction:

The demodulation stage uses a passive low-pass filter to recover the original low-frequency message signal from the modulated pulse waveform. The filter is designed with a nominal cutoff frequency of approximately (100 Hz), subject to the selected resistor and capacitor values. The cutoff frequency is chosen to allow the message component to pass while attenuating unwanted high-frequency switching components and carrier-related harmonics. As the filter suppresses these rapid variations, the pulse waveform becomes smoother and its low-frequency component becomes more prominent. The resulting output is a continuous, approximately sinusoidal waveform that can be compared with the original message signal to evaluate the performance of the modulation and demodulation process. Successful reconstruction depends on the filter response, the selected modulation method, and the characteristics of the signal supplied to the filter. In particular, PWM and PPM signals may require appropriate pulse recovery or conversion circuitry before low-pass filtering.

11. Cost Analysis:

Table 4: Project Cost

S.L No	Apparatus Name	Cost (BDT)
1	IC	650
2	IC Base	170
3	Terminal Block	100
4	PCB Print + Delivery charge	920
5	Circuit Print	400
6	Box	250
7	Socket	300
8	Binding Tape	60
9	Square	10
10	Connector	60
Total		**2920**

12. Simulation Result:

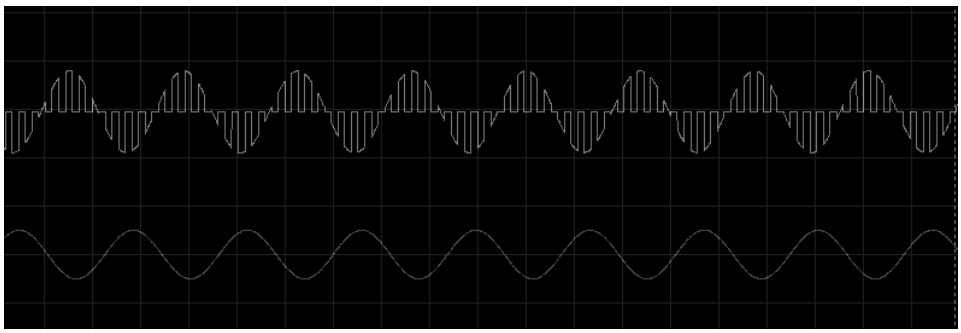


Figure 13: Simulated output of PAM

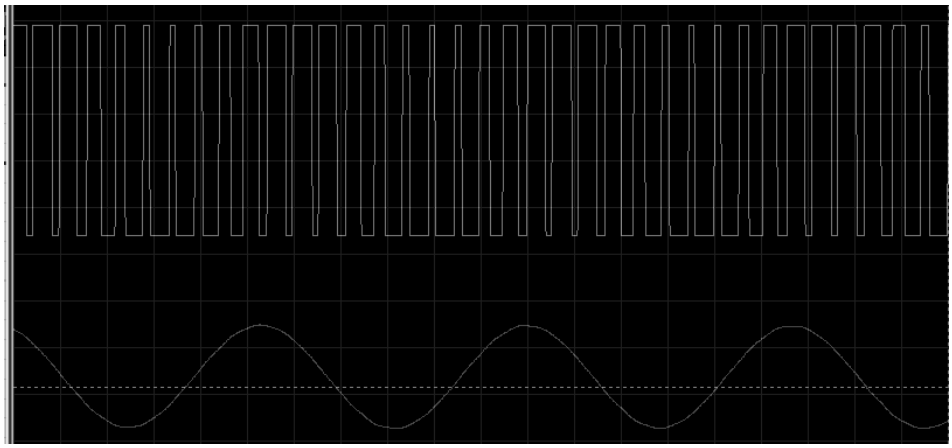


Figure 14: Simulated output of PWM

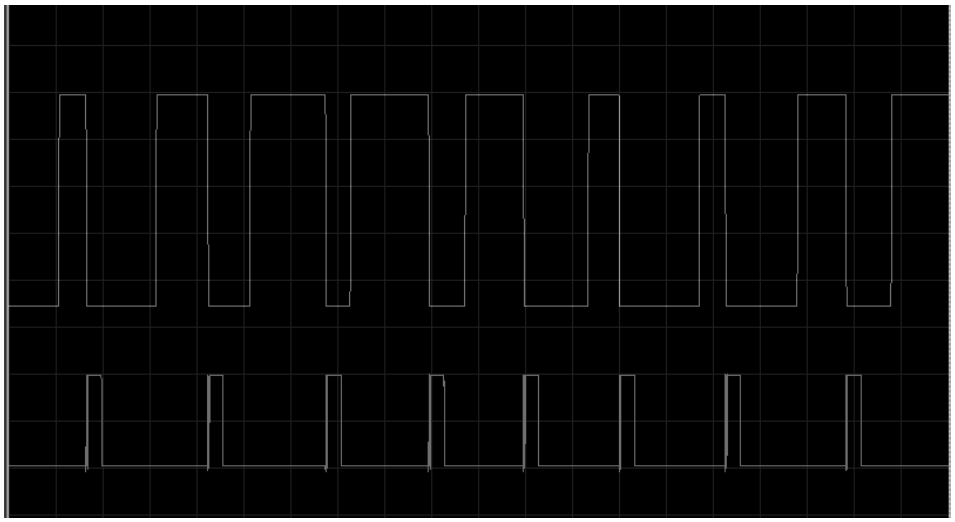


Figure 15: Simulated output of PPM

13. Experimental Result:

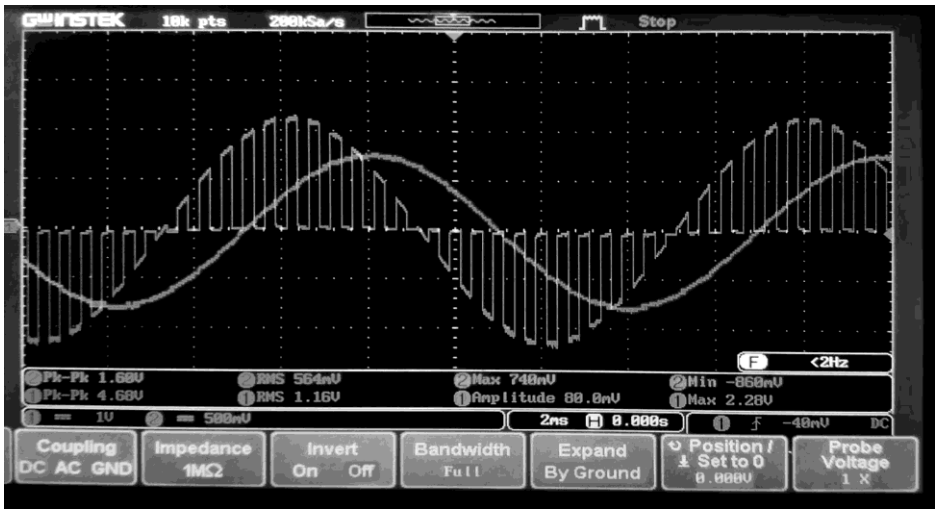


Figure 16: Output shape of PAM

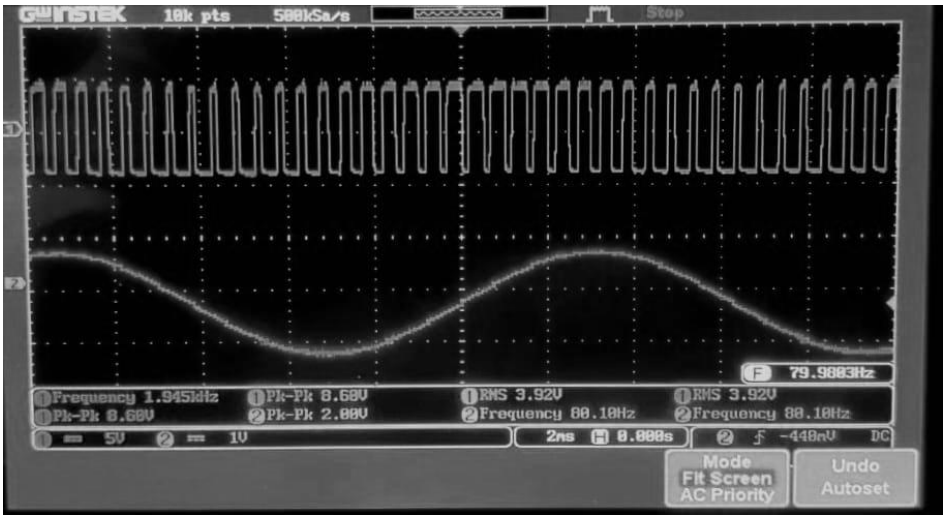


Figure 17: Output shape of PWM

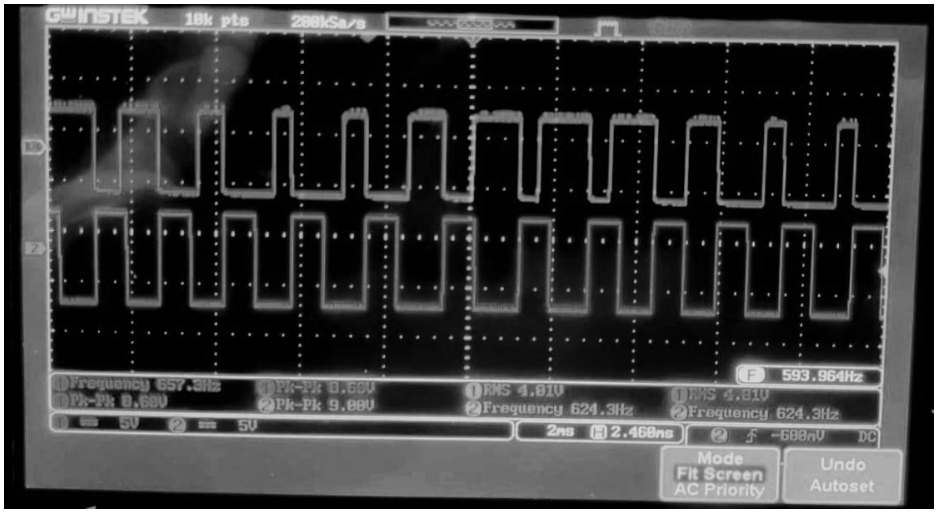


Figure 18: Output shape of PPM

14.Conclusion:

The integrated Pulse Modulation Trainer Kit was successfully designed, prototyped, fabricated, and packaged as a fully functional, low-power educational hardware platform. The project systematically progressed from theoretical circuit calculations and breadboard prototyping to a custom-fabricated Printed Circuit Board (PCB) housed securely inside an enclosure box. All three pulse modulation schemes—Pulse Amplitude Modulation (PAM), Pulse Width Modulation (PWM), and Pulse Position Modulation (PPM)—were experimentally generated and analyzed under strict power supply limits (Dual Supply $\leq \pm 10$V DC) and input frequency constraints (50 Hz $< f_m <$ 95 Hz).

Key engineering challenges, such as correcting inverted PPM pulses through an Op-Amp Unity-Gain Inverting Amplifier stage and maintaining linear operation without signal clipping ($V_m < V_c$), were successfully resolved. Furthermore, the integrated 2nd-order active Sallen-Key Low-Pass Filter demonstrated smooth baseband signal recovery with minimal distortion. Ultimately, this project delivers a robust, standardized, plug-and-play laboratory kit that bridges abstract digital communication theories with practical hardware experience for junior undergraduate engineering students.